

VYSHNAV K KUMAR

FULL STACK DEVELOPER | MERN STACK | GENERATIVE AI & RAG

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PROFESSIONAL SUMMARY

Full Stack Developer with 5+ years of experience designing, developing, and scaling enterprise web applications using React.js, Next.js, Node.js, Express.js, MongoDB, JavaScript, and TypeScript. Strong background in responsive UI development, REST and GraphQL APIs, authentication, microservices, real-time systems, and cloud-based deployments. Hands-on exposure to Generative AI application development, including LLM API integration, prompt engineering, retrieval-augmented generation (RAG), embeddings, vector search, and AI-assisted engineering workflows. Proven technical lead with experience guiding a 9-member team, improving application performance by 40%, reducing environment setup time by 83%, and maintaining 99.9% service availability.

TECHNICAL SKILLS

Frontend	React.js, Next.js, Redux, Redux Toolkit, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Material UI, Responsive Design
Backend & APIs	Node.js, Express.js, REST API Development, GraphQL, WebSocket, JWT, Authentication & Authorization, Microservices
Databases & Caching	MongoDB, MySQL, PostgreSQL, Mongoose, Sequelize, Redis
Generative AI	LLMs, RAG, Prompt Engineering, Embeddings, Vector Search, OpenAI API concepts, Gemini API concepts, LangChain/LlamaIndex concepts, Agentic AI
Cloud & DevOps	Docker, AWS EC2, Azure exposure, CI/CD, GitLab CI/CD, Git, Nginx, Linux, Kubernetes fundamentals
Engineering	System Design, API Integration, Agile/Scrum, Code Review, TDD, Performance Optimization, Postman, Jira

PROFESSIONAL EXPERIENCE

Technical Lead | Infinite Open Source Solutions LLP, Calicut, India | Jan 2024 - Present

- Lead full-stack architecture and delivery of enterprise MERN applications using React.js, Node.js, Express.js, MongoDB, REST APIs, and reusable service-oriented components.
- Design scalable backend APIs, authentication flows, real-time features, and responsive user interfaces while translating product requirements into maintainable technical solutions.
- Guide a cross-functional team of 9 engineers through sprint planning, code reviews, technical decisions, debugging, release management, and production support.
- Integrated AI-assisted and agentic development workflows using Claude Code, Cursor, GitHub Copilot, and ChatGPT for rapid prototyping, refactoring, debugging, test generation, and documentation.
- Contributed to AI-enabled solution design involving LLM-based summarization, structured outputs, document processing, embeddings, vector retrieval, and RAG-oriented workflows for enterprise applications.
- Engineered an automated demo provisioning pipeline that reduced setup time by 83% (90 seconds to 15 seconds), saving more than 10 engineering hours per week.
- Improved application performance by up to 40% through query optimization, API profiling, caching, code-quality standards, and systematic root-cause analysis.
- Maintained 99.9% service availability through Dockerized deployments, CI/CD automation, monitoring, infrastructure hardening, and incident-response practices.

Software Developer | Infinite Open Source Solutions LLP, Calicut, India | Oct 2021 - Dec 2023

- Developed end-to-end product features using React.js, JavaScript, Node.js, Express.js, MongoDB, MySQL, and RESTful APIs for enterprise web platforms.
- Built reusable UI components, modular backend services, validation layers, database integrations, error handling, and secure JWT-based authentication.
- Modernized legacy modules into service-oriented and microservices-based components, improving scalability, maintainability, and deployment flexibility.
- Implemented real-time dashboards and event-driven functionality using WebSocket, Redis, and RabbitMQ-based processing patterns.
- Resolved more than 50 critical production issues through log analysis, performance profiling, database investigation, regression testing, and documented root-cause analysis.
- Collaborated with product managers, UI/UX designers, QA engineers, and DevOps teams to deliver production-ready releases in Agile sprints.

SELECTED PROJECTS

AI-Powered Customer Engagement Platform | *React.js, Node.js, Python/FastAPI, MongoDB, PostgreSQL, LLMs, RAG*

Designed solution components for an AI-enabled customer engagement platform that processes conversations and documents to generate summaries, requirements, searchable knowledge, sentiment insights, and structured business outputs. Worked with LLM integration concepts, prompt workflows, embeddings, vector retrieval, RAG, background queues, and multi-tenant API design.

Network Marketing Enterprise Platform | *React.js, Node.js, Express.js, MongoDB, MySQL, Redis*

Built and scaled a multi-tenant full-stack platform with complex hierarchical user structures, commission engines, financial workflows, genealogy dashboards, authentication, reporting, and high-volume API/database operations.

Learning Management System | *MERN Stack, AWS, WebSocket*

Developed an LMS platform with responsive interfaces, course and user management, student progress tracking, automated assessments, real-time functionality, and analytics dashboards deployed on AWS.

Process Automation & DevOps Pipeline | *Node.js, Docker, GitLab CI/CD, AWS*

Created automated provisioning and deployment workflows that reduced setup time by 83%, improved release consistency, and supported 99.9% service availability across production environments.

EDUCATION & CERTIFICATIONS

B.Tech in Electronics and Communication Engineering | T K M College of Engineering, Kollam, Kerala

Certifications: Penetration Tester - RedTeam Hacker Academy | Full Stack Development - CCBP Tech 4.0 Intensive

ACHIEVEMENTS

- Best R&D Performer Award for product innovation, backend optimization, and engineering improvements.
- Process Optimization Excellence recognition for reducing custom demo setup time by 83% and improving sales enablement.